

Project Name: ConnectHub – Student Networking

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RAID Log

Risk	Description	Impact	Mitigation Strategy
Authentication Failure	Clerk integration may fail in production	High	Regular testing, fallback login plan
Cloudinary Upload Errors	Issues while uploading large files or formats	Medium	Limit file size and validate file types
UI Responsiveness	Mobile layout may break due to dynamic content	Medium	Use TailwindCSS breakpoints & testing
Backend Downtime	Node.js server may crash unexpectedly	High	Use logging, try-catch blocks, Render logs monitoring
API Rate Limits	Cloudinary or Clerk may throttle excessive calls	Low	Optimize API calls, apply caching

Assumption	Impact if False	Contingency Plan
Students will mostly access via mobile	UI may break	Implement responsive design from beginning
Project files are within upload size limit	Upload failure	Compress and validate file size before upload

Issue	Resolution
Login bug in Clerk integration	Adjusted Clerk config and used dynamic routing
Cloudinary not accepting files	Fixed by setting proper API keys and file validation

Dependency	Impact	Management Plan
Clerk (Auth Provider)	Critical	Monitor uptime and have alt auth setup in future
MongoDB Atlas	Medium	Ensure reliable hosting and backup routines

Lessons Learnt Log

Aspect	Lesson Learnt
Planning	A clear WBS helped divide and track progress efficiently.
Frontend	Next.js routing and component reuse saved time and kept the UI modular.
Authentication	Clerk is easy to use but requires proper configuration and callback handling.
Media Management	Cloudinary integration is powerful, but file size/type validation is essential to avoid runtime errors.
Backend	Middleware and proper API structuring in Express.js made debugging easier.
Team Communication	Frequent GitHub commits and task logs helped maintain clarity.
Documentation	Maintaining logs, code comments, and test cases early made final documentation faster.
Testing	Testing each module individually prevented large-scale bugs in deployment.